

ML Ops Assignment 2

ML OPS assignment for Hugging Face Fine-Tuning, Experiment Tracking Model Deployment

Name: Vivekanandhan R

Roll Number: G25AIT2077

1. Introduction & Objective

This project implements the pipeline by fine-tuning a pre-trained **DistilBERT** model on **UCSD Goodreads reviews** to classify books into **eight genres**. The work demonstrates end-to-end MLOps practices: production scripting, experiment tracking, cloud GPU training, evaluation with artifact logging, and model deployment.

Core ML Ops practices demonstrated:

- **Productionizing notebooks:** Modular Python scripts for data loading, training, evaluation, and deployment.
 - **Experiment tracking:** Logging metrics, hyperparameters, and artifacts via **Weights & Biases (W&B)**.
 - **Cloud GPU training:** Leveraging **Kaggle's** free GPU infrastructure.
 - **Model deployment:** Publishing the fine-tuned model to the **Hugging Face Hub**.
 - **Reproducibility:** Scripts and tracked experiments ensure the workflow is repeatable.
-

2. Dataset

Attribute	Value
Source	UCSD Book Graph
Genres	poetry, children, comics_graphic, fantasy_paranormal, history_biography, mystery_thriller_crime, romance, young_adult
Features	Review text only
Size	8 genres × 2,000 reviews sampled = ~16,000 total
Preprocessing	Tokenized with DistilBERT tokenizer; truncated/padded to 512 tokens
Split	800 training / 200 test per genre (80/20 split)

The dataset is loaded via HTTP streaming from UCSD mirrors, sampled to 2,000 reviews per genre for computational efficiency, and split stratified by genre.

3. Model & Training Setup

Model: distilbert-base-cased

Tokenizer: DistilBertTokenizerFast

Classes: 8 genres

Max length: 512 tokens

Device: CUDA (Kaggle T4 x2)

Hyperparameters:

Parameter	Value
Epochs	3
Batch size (train)	10
Batch size (eval)	16
Learning rate	5e-5
Warmup steps	100
Weight decay	0.01
Evaluation strategy	steps
Logging	every 100 steps

DistilBERT (66M parameters) was selected because it is lighter and faster than full BERT while retaining strong performance, making it ideal for educational fine-tuning on limited GPU time.

4. Implementation Highlights

Kaggle Secrets Management

API tokens are stored securely via **Kaggle Secrets** rather than hard-coding:

- `WANDB_API_KEY` — W&B authentication.
- `HF_TOKEN` — Hugging Face Hub write access.

```
from kaggle_secrets import UserSecretsClient
secrets = UserSecretsClient()
WANDB_API_KEY = secrets.get_secret('WANDB_API_KEY')
HF_TOKEN = secrets.get_secret('HF_TOKEN')
```

Weights & Biases Integration

W&B is initialized before training with the full hyperparameter configuration:

```
wandb.init(
    project='mlops-assignment2',
    name='distilbert-goodreads-run-1',
    config={'model': model_name, 'epochs': 3, ...}
)
```

TrainingArguments reports to wandb, and the Trainer streams loss, evaluation metrics, and model checkpoints automatically.

Evaluation Metrics

The custom `compute_metrics` function returns **accuracy** and **weighted F1** to the Trainer:

```
def compute_metrics(pred):
    labels = pred.label_ids
    preds = pred.predictions.argmax(-1)
    acc = accuracy_score(labels, preds)
    f1 = precision_recall_fscore_support(labels, preds, average='weighted')[2].mean()
    return {'accuracy': acc, 'f1': f1}
```

Hugging Face Hub Deployment

The fine-tuned model and tokenizer are pushed to the Hub under the repository `Vivek-ML-Projects/distilbert-goodreads-genres`:

```
from huggingface_hub import login

login(token=HF_TOKEN)

model.push_to_hub("Vivek-ML-Projects/distilbert-goodreads-genres")
tokenizer.push_to_hub("Vivek-ML-Projects/distilbert-goodreads-genres")

wandb.run.summary["huggingface_model"] =
"https://huggingface.co/Vivek-ML-Projects/distilbert-goodreads-genres" "
```

Inference from Hub

A demonstration cell loads the deployed model directly from Hugging Face Hub and performs interactive genre prediction on user-supplied book reviews, confirming the deployment is functional.

5. Evaluation Results

After 3 epochs of fine-tuning, the model was evaluated on the held-out test set.

Metric	Score
Accuracy	0.570
F1 Score	0.586
Eval Loss	2.404

6. Challenges Faced & Learnings

1. **Kaggle notebook import error:** For some reason, uploading the notebooks contents failed.
 - **Resolution:** Imported entire notebook by import option under code section rather than home -> notebooks.
2. **Accelerator Error:** Faced accelerator error while training the model. It's due to the token exhaustion for distilbert which supports maximum of 30552 tokens only
 - **Resolution:** By mistake, I have copied the training model for 2 times in various places. commenting one of the trainer call resolved this problem.

3. **WandB Reporting issue:** First couple of times my execution results were not reported to WandB dashboard.

- **Resolution:** There was a command under training arguments was mentioned as `report_to=[]`, updated this to `report_to='wandb'` and started seeing my dashboard.

7. Conclusion

I'm able to successfully demonstrates the complete ML Ops pipeline on the professor's Goodreads genre-classification task: data preparation, model fine-tuning with live experiment tracking, evaluation with artifact logging, and deployment to a public model repository. All components are reproducible from the tracked W&B configuration and the versioned Hugging Face Hub model.

8. Submission Links

Resource	Public Link
GitHub Repository	https://github.com/VivekanandhanR/MLOPS-Assignment
Kaggle Notebook	https://www.kaggle.com/code/rvivekanandhan/g25ait2077-vivekanandhanr-mlops-assignment-2
Hugging Face Model	https://huggingface.co/Vivek-ML-Projects/distilbert-goodreads-genres
W&B Dashboard	https://wandb.ai/vivek-iitj/mlops-assignment2
